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Lab: Lab 4

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a. Program Description:

The purpose of this program is to read data from a file and make use of lists and tuple and output contents of the file. The information that's calculated from the file is no of lines, characters, words etc.

b. Program Source Code:

```
from print_me_first import *
print_me_time()

def counter(filename, lineList):
    fileContents = open(filename, "r")
    lineList = fileContents.readlines()
    lineCount = len(lineList)
    fileContents.close()
    wordCount = 0
    charCount = 0
    for line in lineList:
        outputLine = line.rstrip("\n")
        words = line.split()
        wordCount += len(words)
        for ch in outputLine:
            if ch:
                charCount += 1
    return(lineCount, wordCount, charCount, lineList)

def countandprint(noOfLines, noOfWords, noOfChar, lineList):
    count = 0
    charUpper = 0
    charLower = 0
    charSpace = 0
    charDigits = 0
    specialChars = [".", "!", "?"]
    charSentences = 0
    for line in lineList:
        outputLine = line.rstrip("\n")
        count += 1
```

```
def countandprint(noOfLines, noOfWords, noOfChar, lineList):
    count = 0
    charUpper = 0
    charLower = 0
    charSpace = 0
    charDigits = 0
    specialChars = [".", "!", "?"]
    charSentences = 0
    for line in lineList:
        outputLine = line.rstrip("\n")
        count += 1
        print("Line  {:3d} : {}".format(count, outputLine))

        for ch in outputLine:
            if ch.isupper():
                charUpper += 1
            if ch.islower():
                charLower += 1
            if ch.isspace():
                charSpace += 1
            if ch.isdigit():
                charDigits += 1
            if ch in specialChars:
                charSentences += 1

    print("Total number of lines: {:5d}".format((noOfLines)))
    print("Total number of words: {:5d}".format(noOfWords))
    print("Total number of characters: {:5d}".format(noOfChar))
    print("Total number of uppercase letters: {:5d}".format(charUpper))
    print("Total number of lowercase letters: {:5d}".format(charLower))

countandprint()
```

```
        charSentences += 1
    print("Total number of lines: {:5d}".format((noOfLines)))
    print("Total number of words: {:5d}".format(noOfWords))
    print("Total number of characters: {:5d}".format(noOfChar))
    print("Total number of uppercase letters: {:5d}".format(charUpper))
    print("Total number of lowercase letters: {:5d}".format(charLower))
    print("Total number of spaces: {:5d}".format(charSpace))
    print("Total number of digits: {:5d}".format(charDigits))
    print("Total number of sentences: {:5d}".format(charSentences))

fileName = "test.txt"
lineList = []

(noOfLines, noOfWords, noOfChar, lineList) = counter(fileName, lineList)

countandprint(noOfLines, noOfWords, noOfChar, lineList)
```

c. Program Output

```
C:\Users\shwetachauhan\AppData\Local\Programs\Python\Python310\python.exe C:\Users\shwetachauhan\PycharmProjects\Course_scripts\filecounter.py
Name      : CNET-142 - Shweta Chauhan Lab file Counter
Program   : filecounter.py
Current Time : Apr-17-2022 Sun (11:45:11PM)
Line 1 : Python is an interpreted, high-level, general-purpose
Line 2 : programming language. Created by Guido van Rossum and
Line 3 : first released in 1991, Python's design philosophy
Line 4 : emphasizes code readability with its notable use of
Line 5 : significant whitespace. Its language constructs and
Line 6 : object-oriented approach aim to help programmers
Line 7 : write clear, logical code for small and large-scale
Line 8 : projects.
Line 9 :
Line 10 : Python is meant to be an easily readable language.
Line 11 : Its formatting is visually uncluttered, and it often
Line 12 : uses English keywords where other languages use
Line 13 : punctuation. Unlike many other languages, it does
Line 14 : not use curly brackets to delimit blocks, and
Line 15 : semicolons after statements are optional.
Total number of lines: 15
Total number of words: 94
Total number of characters: 662
Total number of uppercase letters: 10
Total number of lowercase letters: 538
Total number of spaces: 92
Total number of digits: 4
Total number of sentences: 6
```